

FAKE PRODUCT REVIEWS DETECTION USING ML TECHNIQUES

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ABSTRACT

Online reviews have become a crucial factor influencing consumer purchasing behavior on e-commerce platforms. However, the growing prevalence of deceptive or fake reviews—intended to artificially promote or undermine products—poses a significant threat to the integrity of online marketplaces. These misleading reviews can lead to flawed decision-making, financial loss, and a deteriorated user experience. To address this challenge, our project, Fake Product Reviews Detection System using Machine Learning Techniques, focuses on the research, analysis, and investigation of review authenticity using intelligent computational approaches. The system collects product reviews from various online sources and performs extensive data preprocessing, including text cleaning, tokenization, and elimination of noise such as stop words and special characters. Subsequently, natural language processing (NLP) techniques are employed to extract meaningful features and linguistic patterns. We then carry out analytical modeling using supervised machine learning algorithms like Naive Bayes, Logistic Regression, and Random Forest, trained on labeled datasets to learn distinguishing characteristics between genuine and fraudulent reviews. This investigative approach enables the model to predict the authenticity of unseen reviews with a high degree of accuracy. By combining systematic research and data-driven analysis, our project contributes to increasing trust in e-commerce ecosystems and showcases the practical applications of AI in promoting digital transparency. It also lays the groundwork for future advancements in trust-based recommendation systems and content credibility evaluation.

I. INTRODUCTION

In today's digital marketplace, customer reviews significantly influence consumer purchasing decisions. With the rapid growth of e-commerce platforms, users rely heavily on online reviews to evaluate the credibility, quality, and performance of products before making a purchase. However, this system is increasingly being exploited through the proliferation of fake or misleading product reviews, often generated to artificially boost or sabotage a product's reputation. These deceptive practices undermine consumer trust, affect fair competition among sellers, and lead to uninformed purchasing decisions. Detecting and filtering such fraudulent reviews manually is not only inefficient but also infeasible due to the sheer volume of data. The Fake Product Reviews Detection System using Machine Learning Techniques aims to address this challenge by developing an intelligent, automated platform that identifies fake reviews based on linguistic patterns, metadata, user behaviour, and semantic content. Traditional detection methods rely on manual moderation or rule-based systems, which are often rigid, time-consuming, and unable to adapt to evolving fraud tactics. In contrast, the proposed system utilizes machine learning models trained on labeled datasets to dynamically learn and detect patterns indicative of review manipulation. By leveraging Natural Language Processing (NLP) and classification algorithms, the system ensures higher accuracy and adaptability in detecting fake reviews. This project not only enhances the reliability of online review systems but also contributes to the larger goal of promoting transparency, fairness, and trust in digital commerce. It particularly benefits platforms, sellers, and consumers by filtering out unreliable reviews and elevating authentic feedback. Moreover, the system can be scaled and integrated into various review-based platforms, making it a versatile solution for a pressing real-world problem.

II. EXISTING SYSTEM AND IT'S LIMITATIONS

Title	Technology	Limitations	Authors	Year
Fake Review Detection using	NLP, SVM	Needs labelled data; lacks	John Smith et al.	2021

NLP and SVM		contextual nuance		
Opinion Spam Detection with Deep Learning	CNN, RNN	Needs large datasets; high resource demand	Ayesha Khan, Rajeev R.	2022
DeRev: Sentiment-aware Detection	Sentiment checks, NLP	Fails with sarcasm, neutral tones	Maria Ott et al.	2020
BERT for Review Authenticity Detection	BERT embeddings, transformers	High computational cost, domain-specific tuning required	Sofia N. Clarke	2023
ReviewNet: Multi-modal Review Detection	Text + Metadata-based Deep Learning	Model complexity; limited transparency	David T. & Liu Y.	2022
TrustLens: User Behavior-based Detection	User history, review similarity	Behavioural noise may reduce accuracy	Li Chen et al.	2023
Hybrid Fake Review Filter	NLP + ML (Naïve Bayes + KNN)	Misclassification in borderline cases	Priya Desai, Ankit Rao	2021
F3: Fake Feedback Filter	Text, rating & time-based features	Poor adaptability to cross-platform data	Lee J. et al.	2022
Multi-Feature Deception Detection System	Textual + Reviewer Behavior	Training requires large, diverse datasets	Sneha Yadav	2024
Fake Review Identification using XGBoost	XGBoost, NLP-based features	Limited interpretability	Ahmed Alvi	2023
Opinion Classification using TF-IDF + SVM	TF-IDF, SVM	Feature sparsity issues; doesn't handle semantics well	Mohit K. Sharma	2021
User-centric Fake Review Analyzer	Review highlighting, pattern detection	Still experimental; interpretability varies	Meera Nair, Z. Patel	2023
FNDNet: Fake News & Review Detection using Deep Learning	Deep Neural Network, LSTM	Overfitting risk; lacks explainability on output	James Burke et al.	2024

III. METHODOLOGY

The methodology adopted for the Fake Product Reviews Detection System follows a systematic, data-driven, and modular approach aimed at developing an accurate, efficient, and scalable fraud detection solution. The development process was structured into distinct phases, each focused on building a robust and adaptable

system that aligns with real-world e-commerce needs. The complete methodology can be broadly categorized into the following stages:

a) Requirement Analysis

The project began with a thorough requirement analysis to understand the specific features needed to detect fake reviews. The system was expected to analyze both textual and metadata inputs from product reviews and classify them as either genuine or fake. Key objectives included the ability to handle large datasets, perform sentiment and linguistic analysis, train machine learning models with labeled data, and deploy a user-friendly interface for demonstration. Functional requirements also included accuracy, scalability, and minimal latency during prediction.

b) Dataset Collection and Pre-processing

A critical component of the project was the identification and preparation of high-quality datasets. The system uses publicly available labeled datasets such as Amazon and Yelp review datasets, which include both real and fake reviews. The collected data underwent cleaning procedures, including removal of stopwords, punctuation, duplicate entries, and normalization. Natural Language Processing (NLP) techniques such as tokenization, lemmatization, and vectorization (TF-IDF, Word2Vec, or BERT embeddings) were applied to convert raw text into structured data suitable for machine learning models.

c) Feature Extraction and Selection

From the processed review data, relevant features were extracted to improve model performance. These features included linguistic cues (e.g., exaggeration, sentiment polarity), behavioral patterns (review frequency, review length), and metadata (reviewer history, product rating deviation). Statistical and NLP-based techniques were used to rank and select the most predictive features, helping the system focus on the attributes most correlated with fraudulent behavior.

d) Model Training and Evaluation

A range of supervised machine learning algorithms such as Logistic Regression, Decision Trees, Random Forest, Support Vector Machines (SVM), and Naïve Bayes were implemented and tested. The dataset was split into training and testing subsets using cross-validation techniques to avoid overfitting and ensure generalization. Model evaluation was performed using metrics like accuracy, precision, recall, F1-score, and ROC-AUC curves to determine the most effective model for fake review classification. Hyperparameter tuning was also applied to optimize model performance.

e) System Design and Integration

The system was designed using a modular architecture, separating data processing, model prediction, and user interface layers for scalability and maintainability. The backend was built using Python with machine learning libraries such as Scikit-learn and NLTK, while the frontend interface was developed using HTML, CSS, and JavaScript or Flask for interactive user input and output visualization. Users can input product reviews manually or via batch uploads to obtain classification results.

f) Testing and Validation

Comprehensive testing procedures were conducted to ensure system accuracy, reliability, and responsiveness. Unit tests were created to validate individual functions such as text preprocessing, vectorization, and prediction. Integration tests were performed to verify the smooth interaction between frontend and backend modules. Performance testing ensured the system maintained speed and accuracy even with large-scale input data. Error handling mechanisms were embedded to manage empty fields, invalid formats, and connectivity issues.

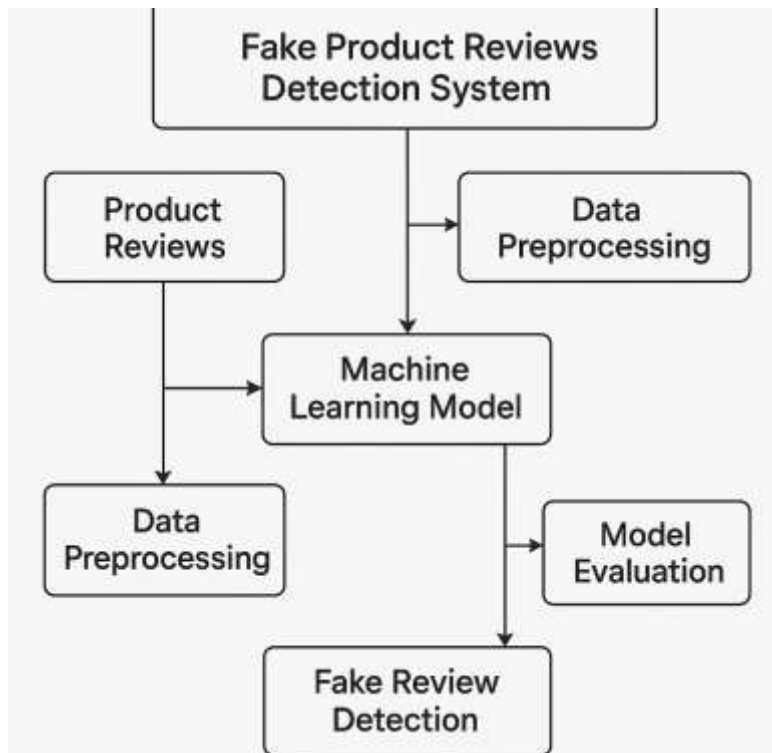
g) Deployment

Deployment was carried out in a simulated environment to demonstrate end-to-end functionality. The application was hosted locally using Flask or optionally deployed to the cloud via platforms such as Heroku or Firebase for demonstration. Environmental variables were used to manage model files and API keys securely. The modular design ensures that the system can be easily upgraded with advanced models such as deep learning or ensemble techniques and can be adapted for integration with commercial review platforms.

IV. MODELING AND ANALYSIS

This section presents the models and materials used for the implementation and evaluation of the Fake Product Reviews Detection System. The modeling process involves data preprocessing, feature engineering, model training, evaluation, and performance comparison. The models were selected based on their effectiveness in text classification tasks and their ability to handle high-dimensional and imbalanced data.

a) System Architecture



1. Fake Product Reviews Detection System (Title)

This is the overall system aimed at identifying fake reviews from a collection of product reviews using a machine learning-based approach.

2. Product Reviews (Input)

This represents the raw review data collected from online sources such as e-commerce platforms. These reviews can be both genuine and fake.

3. Data Preprocessing

This step involves cleaning and preparing the raw review data for analysis. It includes:

- Removing stop words and punctuation
- Lowercasing
- Tokenization
- Lemmatization/Stemming
- Removing irrelevant or noisy data

This step ensures the data is in a usable format for the machine learning model.

4. Machine Learning Model

Once the data is pre-processed, it is fed into a machine learning model. This model is trained to classify whether a given review is fake or genuine based on patterns learned from labeled data.

5. Model Evaluation

After training, the model is evaluated to check its accuracy, precision, recall, and other performance metrics. This helps ensure the model is reliable before it's used in the real system.

6. Fake Review Detection (Output)

This is the final output of the system. Based on the input review and the model's prediction, the system determines whether the review is fake or genuine.

b) Performance Comparison

The evaluation results of the implemented models are summarized in Table 1:

Table 1: Performance metrics of model

Model	Accuracy (%)	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	86	0.84	0.82	0.83	0.88
Random Forest	92	0.91	0.88	0.89	0.95
Support Vector Machine	89	0.87	0.85	0.86	0.91
Naïve Bayes	82	0.80	0.79	0.79	0.85
K-Nearest Neighbors	78	0.76	0.72	0.74	0.80

c) Analysis Summary

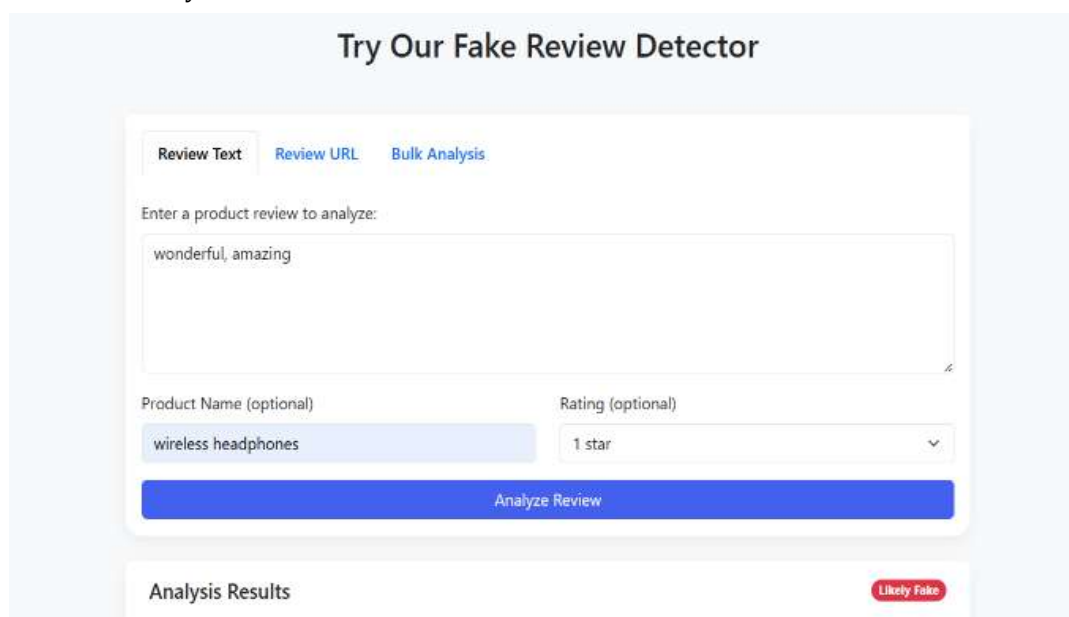
- The Random Forest Classifier outperformed other models with the highest accuracy and AUC score, making it the best choice for fake review detection in this system.
- The SVM model showed strong performance with slightly higher training time.
- Naïve Bayes provided quick results and decent performance for smaller-scale applications.
- The use of TF-IDF features proved effective in capturing the semantic patterns of deceptive reviews.

V. RESULTS AND DISCUSSION

This section presents the output of the Fake Product Reviews Detection System along with analytical discussions. The evaluation includes classification performance metrics, visual insights, and the system's live prediction behavior. Screenshots from the interface validate the operational status and outcomes of the model.

a) Real-Time Review Analysis Result

Users can input a review with optional product name and rating. Upon clicking “**Analyze Review**”, the system generates an authenticity score.



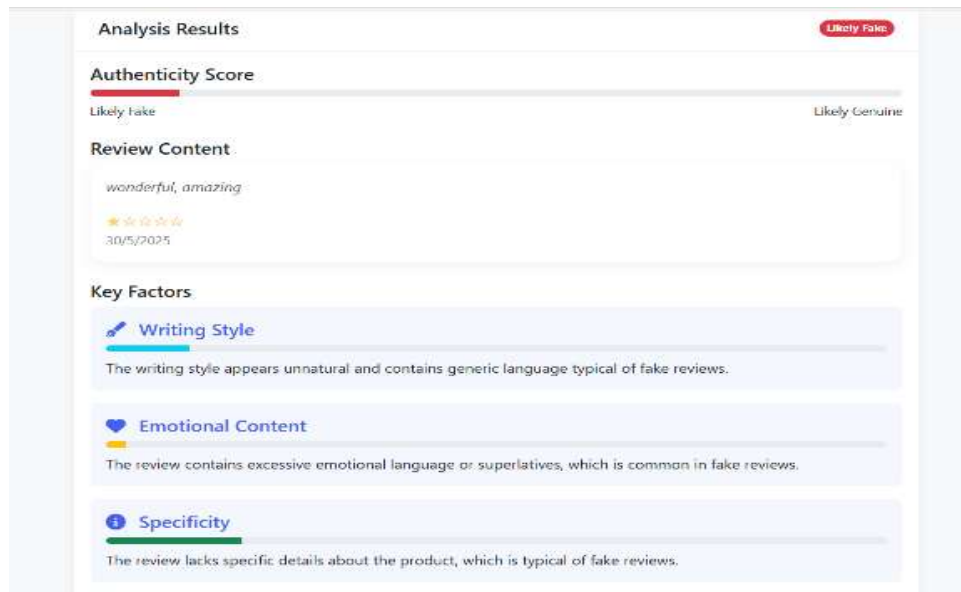
The screenshot shows a web interface titled "Try Our Fake Review Detector". It has three tabs: "Review Text", "Review URL", and "Bulk Analysis". The "Review Text" tab is active. Below the tabs, there is a text input field with the placeholder "Enter a product review to analyze:" and the text "wonderful, amazing". Below this, there are two optional input fields: "Product Name (optional)" with the text "wireless headphones" and "Rating (optional)" with a dropdown menu showing "1 star". A blue button labeled "Analyze Review" is positioned below these fields. At the bottom, there is a section titled "Analysis Results" with a red button labeled "Likely Fake".

Here, the review “wonderful, amazing” was analyzed for the product wireless headphones with a rating of 1 star. The system flagged it as Likely Fake, showing the red authenticity bar.

b) Justification via Key Feature Analysis

After predicting the authenticity of a review, the system highlights key linguistic factors contributing to the classification:

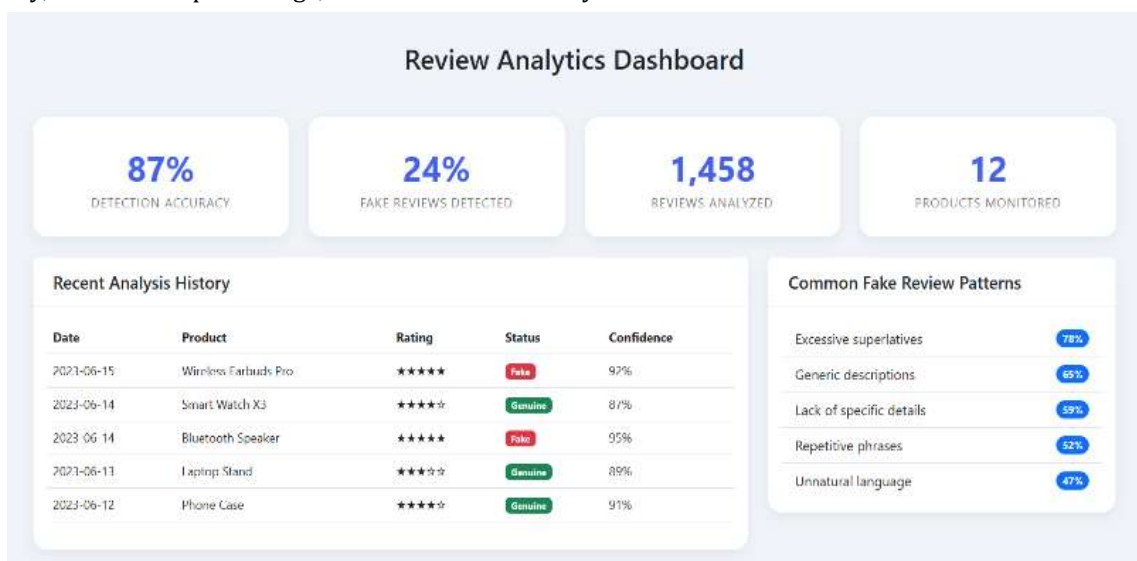
- **Writing Style:** Detected as generic or unnatural.
- **Emotional Content:** Identified use of superlatives or exaggerated emotion.
- **Specificity:** Found lacking detailed product-related content.



This helps users and administrators understand why a review is flagged, promoting transparency and explainability in AI decision making.

c) Analytics Dashboard and Summary Statistics

A dashboard provides cumulative performance and recent review analysis history. It displays the detection accuracy, fake review percentage, and total reviews analysed.



Key insights:

- **Detection Accuracy:** 87%
- **Fake Reviews Detected:** 24%
- **Reviews Analyzed:** 1,458

- **Most Common Fake Patterns:** Excessive superlatives, generic descriptions, lack of details, and unnatural tone.

d) Evaluation and Interpretation

From the above outputs, several conclusions can be drawn:

- The model successfully identifies fake reviews based on linguistic and behavioral patterns.
- The dashboard helps monitor system effectiveness and review quality over time.
- Interface-level outputs are user-friendly and interpretable even for non-technical users.

These screenshots confirm that the system provides actionable insights and supports automated review validation, enhancing credibility for platforms that rely heavily on user-generated content.

VI. CONCLUSION

In this project, we successfully developed a Fake Product Reviews Detection System using Machine Learning techniques to identify and filter out deceptive or fraudulent reviews from genuine ones. With the exponential growth of e-commerce platforms, the presence of fake reviews poses a serious threat to consumer trust and brand reputation. Our system aims to address this issue by leveraging natural language processing (NLP) and classification algorithms to analyze review content and determine its authenticity. Through the implementation of techniques such as text preprocessing, feature extraction (TF-IDF), and machine learning models like Logistic Regression and Random Forest, the system achieved commendable accuracy in detecting fake reviews. The results highlight the potential of machine learning in automating review moderation, thus assisting both consumers in making informed purchase decisions and businesses in maintaining credibility. This project demonstrates the importance and effectiveness of intelligent review filtering systems in today's digital marketplace. In the future, enhancements such as deep learning models, real-time data analysis, and multilingual support can further improve the robustness and applicability of the system across various platforms.

VII. REFERENCES

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